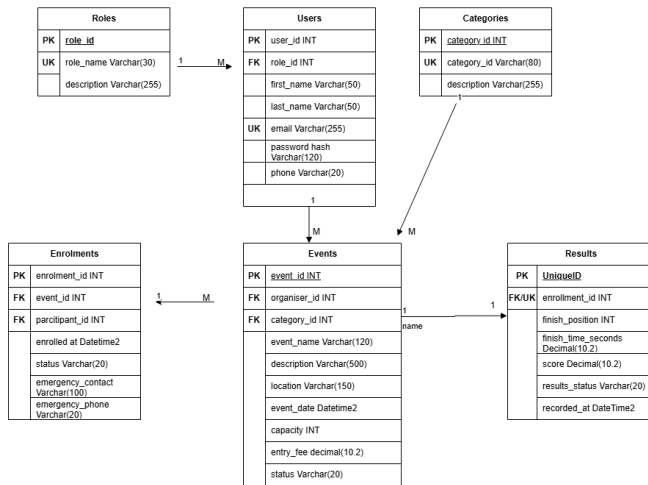


ST10487284

Omphemetse Mabone

Programming 6212

Section A



SECTION B

HTT P Meth od	Route	Descript ion	Role Require d	Request Body	Expected Respons e
POST	/Api/auth/register	Register a new user account	Public	{email, password, first Name, last Name, phone Number, dateOfBirth, role }	201 Created / 400 Bad Request / 422 Unprocessable Entity
POST	/Api/auth/login	Authenticate user and return JWT token	Public	{email, password}	200 OK / 401 Unauthorized / 404 Not Found
POST	/Api/auth/logout	Logout the authenticated user	Any authenticated user	None	200 OK / 401 Unauthorized
GET	/Api/auth/me	Get current authenticated user profile	Any authenticated user	None	200 OK / 401 Unauthorized
GET	/Api/users/{userId}	Get user profile by ID	Any authenticated user	None	200 OK / 401 Unauthorized / 404 Not Found
PUT	/Api/users/{userId}	Update user profile	Any authenticated user	{first Name, last Name, phoneNumbr, dateOfBirth }	200 OK / 400 Bad Request / 401 Unauthorized / 404

					Not Found
PUT	/Api/users/{userId}/enrollments	Get all enrollments for a user	Any authenticated user	{ currentPassword, newPassword }	200 OK / 401 Unauthorized / 404 Not Found
GET	/Api/users/{userId}/enrollments	Get all enrollments for a user	Any authenticated user	None	200 OK / 401 Unauthorized
POST	/Api/events	Create a new event	Organizer	Event details	201 Created / 400 Bad Request / 403 Forbidden
GET	/Api/events	Get all events with filters	Public	Query parameters	200 OK / 400 Bad Request
GET	/Api/events/{eventId}	Get event details	Public	None	200 OK / 404 Not Found
PUT	/Api/events/{eventId}	Update event details	Organizer	Event details	200 OK / 403 Forbidden / 404 Not Found
DELETE	/Api/events/{eventId}	Delete an event	Organizer	None	200 OK / 403 Forbidden / 404 Not Found / 409 Conflict
POST	/Api/events/{eventId}/categories	Add category to an event	Organizer	Category details	201 Created / 403 Forbidden / 404 Not Found /

					409 Conflict
GET	/api/events/{eventId}/categories	Get categories for an event	Public	None	200 OK / 404 Not Found
DELETE	/Api/events/{eventId}/categories/{categoryId}	Remove category from event	Organizer	None	200 OK / 403 Forbidden / 404 Not Found / 409 Conflict
GET	/Api/categories	Get all available categories	Public	None	200 OK
POST	/Api/categories	Create a new category	Organizer	Category details	201 Created / 400 Bad Request / 409 Conflict
GET	/Api/categories/{categoryId}	Get category details	Public	None	200 OK / 404 Not Found
POST	/Api/events/{eventId}/enroll	Enroll participant in event category	Participant	Enrollment details	201 Created / 400 Bad Request / 401 Unauthorized / 404 Not Found

SECTION C

/*

RaceDay System - Part 1 SQL Server Database Script

Compatible with Microsoft SQL Server / SSMS.

Entities:

Roles, Users, Categories, Events, Enrolments, Results

Roles:

1. Organiser

2. Participant

***/**

IF DB_ID('RaceDayDB') IS NULL

BEGIN

CREATE DATABASE RaceDayDB;

END;

GO

USE RaceDayDB;

GO

-- Drop tables in dependency order so the script can be re-run during testing.

```
IF OBJECT_ID('dbo.Results', 'U') IS NOT NULL DROP TABLE dbo.Results;
IF OBJECT_ID('dbo.Enrolments', 'U') IS NOT NULL DROP TABLE dbo.Enrolments;
IF OBJECT_ID('dbo.Events', 'U') IS NOT NULL DROP TABLE dbo.Events;
IF OBJECT_ID('dbo.Categories', 'U') IS NOT NULL DROP TABLE dbo.Categories;
IF OBJECT_ID('dbo.Users', 'U') IS NOT NULL DROP TABLE dbo.Users;
IF OBJECT_ID('dbo.Roles', 'U') IS NOT NULL DROP TABLE dbo.Roles;
GO
```

```
CREATE TABLE dbo.Roles
```

```
(
    role_id INT IDENTITY(1,1) NOT NULL,
    role_name VARCHAR(30) NOT NULL,
    description VARCHAR(255) NULL,

    CONSTRAINT PK_Roles PRIMARY KEY (role_id),
    CONSTRAINT UQ_Roles_role_name UNIQUE (role_name)
);
GO
```

```
CREATE TABLE dbo.Users
```

```
(
    user_id INT IDENTITY(1,1) NOT NULL,
    role_id INT NOT NULL,
    first_name VARCHAR(50) NOT NULL,
    last_name VARCHAR(50) NOT NULL,
    email VARCHAR(120) NOT NULL,
```

```
password_hash VARCHAR(255) NOT NULL,  
phone VARCHAR(20) NULL,  
is_active BIT NOT NULL CONSTRAINT DF_Users_is_active DEFAULT (1),  
created_at DATETIME2 NOT NULL CONSTRAINT DF_Users_created_at DEFAULT  
(SYSDATETIME()),
```

```
CONSTRAINT PK_Users PRIMARY KEY (user_id),  
CONSTRAINT UQ_Users_email UNIQUE (email),  
CONSTRAINT FK_Users_Roles FOREIGN KEY (role_id)  
REFERENCES dbo.Roles(role_id)  
);  
GO
```

```
CREATE TABLE dbo.Categories
```

```
(  
    category_id INT IDENTITY(1,1) NOT NULL,  
    category_name VARCHAR(80) NOT NULL,  
    description VARCHAR(255) NULL,  
  
    CONSTRAINT PK_Categories PRIMARY KEY (category_id),  
    CONSTRAINT UQ_Categories_category_name UNIQUE (category_name)  
);  
GO
```

```
CREATE TABLE dbo.Events
```

```
(
```

event_id INT IDENTITY(1,1) NOT NULL,
organizer_id INT NOT NULL,
category_id INT NOT NULL,
event_name VARCHAR(120) NOT NULL,
description VARCHAR(500) NULL,
location VARCHAR(150) NOT NULL,
event_date DATETIME2 NOT NULL,
capacity INT NOT NULL,
entry_fee DECIMAL(10,2) NOT NULL CONSTRAINT DF_Events_entry_fee DEFAULT
(0.00),
status VARCHAR(20) NOT NULL CONSTRAINT DF_Events_status DEFAULT ('Open'),
created_at DATETIME2 NOT NULL CONSTRAINT DF_Events_created_at DEFAULT
(SYSDATETIME()),

CONSTRAINT PK_Events PRIMARY KEY (event_id),
CONSTRAINT FK_Events_Organizer FOREIGN KEY (organizer_id)
REFERENCES dbo.Users(user_id),
CONSTRAINT FK_Events_Category FOREIGN KEY (category_id)
REFERENCES dbo.Categories(category_id),
CONSTRAINT CK_Events_capacity CHECK (capacity > 0),
CONSTRAINT CK_Events_entry_fee CHECK (entry_fee >= 0),
CONSTRAINT CK_Events_status CHECK (status IN ('Draft', 'Open', 'Closed',
'Completed', 'Cancelled'))
);
GO

CREATE TABLE dbo.Enrolments

```

(
    enrolment_id INT IDENTITY(1,1) NOT NULL,
    event_id INT NOT NULL,
    participant_id INT NOT NULL,
    enrolled_at DATETIME2 NOT NULL CONSTRAINT DF_Enrolments_enrolled_at
    DEFAULT (SYSDATETIME()),
    status VARCHAR(20) NOT NULL CONSTRAINT DF_Enrolments_status DEFAULT
    ('Confirmed'),
    emergency_contact VARCHAR(100) NULL,
    emergency_phone VARCHAR(20) NULL,

    CONSTRAINT PK_Enrolments PRIMARY KEY (enrolment_id),
    CONSTRAINT FK_Enrolments_Event FOREIGN KEY (event_id)
        REFERENCES dbo.Events(event_id),
    CONSTRAINT FK_Enrolments_Participant FOREIGN KEY (participant_id)
        REFERENCES dbo.Users(user_id),
    CONSTRAINT UQ_Enrolments_EventParticipant UNIQUE (event_id, participant_id),
    CONSTRAINT CK_Enrolments_status CHECK (status IN ('Pending', 'Confirmed',
    'Cancelled', 'Completed'))
);
GO

```

```

CREATE TABLE dbo.Results

```

```

(
    result_id INT IDENTITY(1,1) NOT NULL,
    enrolment_id INT NOT NULL,
    finish_position INT NULL,

```

```
finish_time_seconds DECIMAL(10,2) NULL,  
score DECIMAL(10,2) NULL,  
result_status VARCHAR(20) NOT NULL CONSTRAINT DF_Results_status DEFAULT  
('Recorded'),  
recorded_at DATETIME2 NOT NULL CONSTRAINT DF_Results_recorded_at DEFAULT  
(SYSDATETIME()),
```

```
CONSTRAINT PK_Results PRIMARY KEY (result_id),  
CONSTRAINT UQ_Results_enrolment UNIQUE (enrolment_id),  
CONSTRAINT FK_Results_Enrolment FOREIGN KEY (enrolment_id)  
REFERENCES dbo.Enrolments(enrolment_id),  
CONSTRAINT CK_Results_position CHECK (finish_position IS NULL OR  
finish_position > 0),  
CONSTRAINT CK_Results_time CHECK (finish_time_seconds IS NULL OR  
finish_time_seconds > 0),  
CONSTRAINT CK_Results_score CHECK (score IS NULL OR score >= 0),  
CONSTRAINT CK_Results_status CHECK (result_status IN ('Recorded',  
'Disqualified', 'Pending'))  
);  
GO
```

-- Seed roles: exactly the two application roles required by the brief.

```
INSERT INTO dbo.Roles (role_name, description)  
VALUES  
('Organiser', 'Creates and manages RaceDay events and records results'),  
('Participant', 'Registers for RaceDay events and views personal results');  
GO
```

-- Seed users: at least two organisers and two participants.

-- Passwords are sample hashes/placeholders for the planning database only.

INSERT INTO dbo.Users

(role_id, first_name, last_name, email, password_hash, phone)

VALUES

((SELECT role_id FROM dbo.Roles WHERE role_name = 'Organiser'),

**'Lerato', 'Mokoena', 'lerato.organiser@raceday.test', 'HASH_SAMPLE_001',
'0711111111'),**

((SELECT role_id FROM dbo.Roles WHERE role_name = 'Organiser'),

**'Thabo', 'Dlamini', 'thabo.organiser@raceday.test', 'HASH_SAMPLE_002',
'0722222222'),**

((SELECT role_id FROM dbo.Roles WHERE role_name = 'Participant'),

**'Omphemetse', 'Mabone', 'omphemetse.participant@raceday.test',
'HASH_SAMPLE_003', '0733333333'),**

((SELECT role_id FROM dbo.Roles WHERE role_name = 'Participant'),

**'Naledi', 'Molefe', 'naledi.participant@raceday.test', 'HASH_SAMPLE_004',
'0744444444');**

GO

-- Event categories.

INSERT INTO dbo.Categories (category_name, description)

VALUES

('Road Race', 'Road-running events'),

('Trail Run', 'Off-road running events'),

('Cycling', 'Cycling and road cycling events');

GO

-- At least three events.

INSERT INTO dbo.Events

**(organizer_id, category_id, event_name, description, location, event_date, capacity,
entry_fee, status)**

VALUES

(

(SELECT user_id FROM dbo.Users WHERE email = 'lerato.organiser@raceday.test'),

(SELECT category_id FROM dbo.Categories WHERE category_name = 'Road Race'),

'Johannesburg City 10K',

'A 10 kilometre city road race.',

'Johannesburg, Gauteng',

'2026-10-10 07:00:00',

500,

120.00,

'Open'

),

(

(SELECT user_id FROM dbo.Users WHERE email = 'thabo.organiser@raceday.test'),

(SELECT category_id FROM dbo.Categories WHERE category_name = 'Trail Run'),

'Magaliesberg Trail Challenge',

'A scenic trail-running challenge.',

'Magaliesberg, Gauteng',

'2026-11-14 06:30:00',

250,

180.00,

```
'Open'
),
(
(SELECT user_id FROM dbo.Users WHERE email = 'lerato.organiser@raceday.test'),
(SELECT category_id FROM dbo.Categories WHERE category_name = 'Cycling'),
'Pretoria Cycle Classic',
'A community cycling event.',
'Pretoria, Gauteng',
'2026-12-05 06:00:00',
400,
220.00,
'Open'
);
GO
```

-- Enrolments.

```
INSERT INTO dbo.Enrolments
```

```
    (event_id, participant_id, status, emergency_contact, emergency_phone)
```

```
VALUES
```

```
(
(SELECT event_id FROM dbo.Events WHERE event_name = 'Johannesburg City 10K'),
(SELECT user_id FROM dbo.Users WHERE email =
'omphemetse.participant@raceday.test'),
'Confirmed', 'Kagiso Mabone', '0755555555'
),
(
```

```

(SELECT event_id FROM dbo.Events WHERE event_name = 'Johannesburg City 10K'),
(SELECT user_id FROM dbo.Users WHERE email = 'naledi.participant@raceday.test'),
'Confirmed', 'Mpho Molefe', '0766666666'
),
(
(SELECT event_id FROM dbo.Events WHERE event_name = 'Magaliesberg Trail
Challenge'),
(SELECT user_id FROM dbo.Users WHERE email =
'omphemetse.participant@raceday.test'),
'Confirmed', 'Kagiso Mabone', '0755555555'
);
GO

```

-- Results for completed/recorded enrolments.

```

INSERT INTO dbo.Results
    (enrolment_id, finish_position, finish_time_seconds, score, result_status)
VALUES
(
(SELECT e.enrolment_id
    FROM dbo.Enrolments e
    JOIN dbo.Events ev ON ev.event_id = e.event_id
    JOIN dbo.Users u ON u.user_id = e.participant_id
    WHERE ev.event_name = 'Johannesburg City 10K'
    AND u.email = 'omphemetse.participant@raceday.test'),
12, 3245.50, 88.00, 'Recorded'
),

```

```
(
(SELECT e.enrolment_id
FROM dbo.Enrolments e
JOIN dbo.Events ev ON ev.event_id = e.event_id
JOIN dbo.Users u ON u.user_id = e.participant_id
WHERE ev.event_name = 'Johannesburg City 10K'
AND u.email = 'naledi.participant@raceday.test'),
18, 3412.25, 82.00, 'Recorded'
);
GO
```

-- Verification queries for SSMS.

SELECT * FROM dbo.Roles;

Results		Messages	
	role_id	role_name	description
1	1	Organiser	Creates and manages RaceDay events and records r...
2	2	Participant	Registers for RaceDay events and views personal res...

SELECT * FROM dbo.Users;

Results

Messages

	user_id	role_id	first_name	last_name	email	password_hash	phone	is_active	created_at
1	1	1	Lerato	Mokoena	lerato.organiser@raceday.test	HASH_SAMPLE_001	0711111111	1	2026-09-04 20:14:27.4346168
2	2	1	Thabo	Dlamini	thabo.organiser@raceday.test	HASH_SAMPLE_002	0722222222	1	2026-09-04 20:14:27.4346168
3	3	2	Omphemetse	Mabone	omphemetse.participant@raceday.test	HASH_SAMPLE_003	0733333333	1	2026-09-04 20:14:27.4346168
4	4	2	Naledi	Molefe	naledi.participant@raceday.test	HASH_SAMPLE_004	0744444444	1	2026-09-04 20:14:27.4346168

SELECT * FROM dbo.Categories;

Results		Messages	
	category_id	category_name	description
1	1	Road Race	Road-running events
2	2	Trail Run	Off-road running events
3	3	Cycling	Cycling and road cycling events

SELECT * FROM dbo.Events;

Results Messages											
	event_id	organizer_id	category_id	event_name	description	location	event_date	capacity	entry_fee	status	created_at
1	1	1	1	Johannesburg City 10K	A 10 kilometre city road race.	Johannesburg, Gauteng	2026-10-10 07:00:00.0000000	500	120.00	Open	2026-09-04 20:15:46.4131447
2	2	2	2	Magaliesberg Trail Challenge	A scenic trail-running challenge.	Magaliesberg, Gauteng	2026-11-14 06:30:00.0000000	250	180.00	Open	2026-09-04 20:15:46.4131447
3	3	1	3	Pretoria Cycle Classic	A community cycling event.	Pretoria, Gauteng	2026-12-05 06:00:00.0000000	400	220.00	Open	2026-09-04 20:15:46.4131447

SELECT * FROM dbo.Enrolments;

Results Messages								
	enrolment_id	event_id	participant_id	enrolled_at	status	emergency_contact	emergency_phone	
1	1	1	3	2026-09-04 20:16:02.7055563	Confirmed	Kagiso Mabone	0755555555	
2	2	1	4	2026-09-04 20:16:02.7055563	Confirmed	Mpho Molefe	0766666666	
3	3	2	3	2026-09-04 20:16:02.7055563	Confirmed	Kagiso Mabone	0755555555	

SELECT * FROM dbo.Results;

Results Messages								
	result_id	enrolment_id	finish_position	finish_time_seconds	score	result_status	recorded_at	
1	1	1	12	3245.50	88.00	Recorded	2026-09-04 20:16:02.7245666	
2	2	2	18	3412.25	82.00	Recorded	2026-09-04 20:16:02.7245666	

GO